CHAPTER EIGHT

Assessing children's spatial thinking: Insights, challenges, and implications

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Abstract

In the past few decades, interest in children's spatial thinking has increased substantially, and consequently, interest in spatial assessments for children has also increased. However, there are not many reliable, validated, and widely accessible spatial assessments

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for this segment of the population, which affects researchers' ability to conduct and interpret spatial thinking research. While some limitations of these tests relate to broader issues with spatial assessments in general (see Uttal et al., 2024), creating assessments that are appropriate for children presents unique challenges. In this chapter, we review the current state of tests of children's spatial thinking, including mental rotation and perspective-taking. We draw on insights from psychometrics, open science, and cognitive development research. Furthermore, we examine how spatial assessments affect research on the relation between spatial and STEM abilities, particularly research aimed at leveraging spatial thinking through interventions and training that improve children's spatial skills and, in turn, their STEM performance (i.e., Bruce & Hawes, 2015; Cheng & Mix, 2014; Hawes et al., 2022; Judd & Klingberg, 2021; Mix et al., 2021). Lastly, we outline recommendations for improving these assessments to ultimately improve research and theory creation on the development of spatial thinking.

Keywords: Assessment and measurement, Development, Psychometrics, Spatial thinking

Spatial thinking is an important part of everyday life and learning. In addition to being essential for tasks like navigation or arranging objects in a room, spatial thinking is particularly important for STEM disciplines. This relationship has been explored broadly (Uttal & Cohen, 2012) and within specific STEM domains, including, but not limited to, mathematics (Atit et al., 2022), chemistry (Harle & Towns, 2011), anatomy (Roach et al., 2021), and geoscience (Nazareth et al., 2019). Importantly, spatial thinking has been shown to be a significant and unique contributor to STEM achievement and participation, even when controlling for verbal and mathematical abilities (Shea et al., 2001; Wai et al., 2009). More recently, research has focused on training and intervention studies aimed at improving students' STEM abilities directly and indirectly by enhancing their spatial skills (Cheng & Mix, 2014; Langlois et al., 2020; Sorby et al., 2013). This research on spatial training is supported by robust findings regarding the malleability of spatial skills (Uttal et al., 2013). Despite this emphasis on leveraging spatial thinking for STEM education, many unanswered questions remain about the structure and framework of the skills that are in this domain (Uttal et al., 2024). Moreover, efforts to delineate the various factors that comprise spatial thinking have yielded different results (Hegarty & Waller, 2005).

1. The need for developmental spatial thinking research

In addition to unanswered questions about the structure of spatial thinking, many questions remain about the development of spatial skills. For example, different studies that have examined the developmental trajectory of spatial transformation abilities have found different answers regarding when children demonstrate such abilities (Frick et al., 2014a). Typically, spatial skills assessed in adults have been evaluated in children with minimal changes to the assessments. Thus, the lack of consensus about how different spatial skills relate to each other in research with adults is echoed in research with children. However, focusing on developmental research can be key in unraveling the structure of spatial thinking; studying and comparing the development of various spatial abilities may provide insight into how different spatial skills are related. Nevertheless, research with children presents unique challenges. Creating and administering assessments for children can be challenging, and there are fewer spatial assessments for children available compared to adults.

1.1 The importance of assessment and construct validity

To understand how the lack of rigorous assessments for children may affect research on spatial thinking, we must consider the importance of assessment in training and intervention research, as well as the current state of spatial thinking assessment. Operationalizing constructs is essential for quality psychological research. In cognitive psychology, this operationalization typically occurs through the creation and use of assessments targeting specific cognitive constructs. However, in the case of spatial thinking, the development of constructs has often followed from assessments (Hegarty & Waller, 2005); many spatial assessments were designed to meet specific needs rather than to provide rigorous measures of well-defined constructs. Furthermore, many studies measuring spatial skills rely on only one or two specific types of spatial skills, yet they often refer to these as general measures of spatial thinking.

Given that assessments play a critical role in measuring spatial thinking and determining which spatial skills are examined, it is essential to understand what the assessments we use actually measure. Generally, researchers separate spatial thinking into small-scale spatial thinking, which deals with smaller objects and spaces, and large-scale spatial thinking, which focuses on navigation and wayfinding. There are more small-scale spatial assessments than large-scale ones. Uttal et al., (2024) recently published a review addressing current challenges with spatial thinking tests. Among these challenges are difficulties in understanding what different spatial tests measure and how they relate to one another. This can leave researchers uncertain about which tests are most appropriate for contexts such as investigating spatial training for STEM improvement. While these issues have been addressed in assessments designed for adults (Uttal et al., 2024), spatial assessments for children are similarly affected. Additionally, assessments developed for children, particularly young children, present unique challenges that further complicate understanding and interpreting these tests.

2. What spatial assessments are there for young children?

To understand the current state of spatial assessment in children, we need to examine which tests are being used. While there are too many existing tests to review all of them (Schenck & Nathan, 2024), some assessments are more commonly used than others. Here, we will review some of the most common spatial assessments for children and use these exemplar tests to highlight and discuss broader issues related to children's spatial testing. To identify the tests available for young children, we must define the age range that "young children" encompasses. On the lower end, children younger than 2 or 3 years old are typically given assessments that resemble infant assessments more than adult assessments. Additionally, while some judgment-based spatial assessments are administered to three-year-olds, there are far fewer tests for younger children compared to older ones (Verdine et al., 2017). On the upper end of the age range, researchers have administered adult spatial tests to children as young as 7–8 years old (Hoyek et al., 2012). Thus, in this chapter, we will generally review assessments designed for children between the ages of 3- and 8-years-old. Tables 1, 2, 3 and 4 provide examples of different types of spatial assessments for children.

2.1 Mental rotation

As mentioned above, mental rotation is the most used type of spatial thinking test. Mental rotation is often defined as the cognitive operation of transforming a mental image by imagining its rotation through different orientations in space. This definition stems from the original mental rotation test by Shepard and Metzler (1971), in which participants determined whether two 3D cube figures presented at different orientations were the same or different. In addition to measuring accuracy, participants' reaction times were also recorded. Reaction time was strongly and linearly related

Table 1 Examples of tests used to assess children's mental rotation.

Test Name	Source	Original age range (years old)
Picture Rotation Test	Quaiser-Pohl, 2003	4–6
Children's Mental Transformation Test[a]	Levine et al., 1999	4–7
Ghost Mental Rotation Test	Frick et al., 2013	3–5 (though 3-year-olds were near chance level)
MRT-Animal	Titze et al., 2010 (based on Vandenburg & Kuse, 1978)	8–10
Gilligan et al., 2019	Gilligan et al., 2019 (modified from Broadbent et al., 2014)	6–10

[a] The original CMTT was not solely a mental rotation test but a general mental transformation test. However, it has been used as a substitute for mental rotation tests several times and also modified to include only rotation items.

to the angular disparity between the two cube figures. Some researchers suggest that this linear relationship indicates that participants were imagining the rotation of the cube figures. in an analog process; the more they had to rotate the cube figures., the longer it took. Thus, many researchers have assumed that both in the original mental rotation task and in the numerous subsequent variations of the test individuals were mentally rotating the test stimuli (i.e., Just & Carpenter, 1985; Khooshabeh et al., 2013). However, evidence suggests that individuals do not always imagine rotation on these tests. Several studies have shown that individuals can instead use alternative strategies, such as analytic strategies that are not spatial in nature (Boone & Hegarty, 2017; Hegarty, 2018; Just & Carpenter, 1985).

The study of children's mental rotation dates back to the 1970s. Marmor (1975, 1977) investigated when children first represent movement in mental images, a concept known as kinetic mental imagery at the time. Piaget and Inhelder (1971) proposed that kinetic imagery, or mental imagery that represents movement, emerges during the concrete operational stage, which occurs around ages 7–8. Marmor (1975, 1977) used Shepard and Metzler's mental rotation paradigm but found the original 3D cube figure stimuli too difficult for young children and redesigned the task to be appropriate for 4- and 5-year-olds (see Fig. 1). Marmor found that both 4- and 5-year-olds seemed able to perform mental imagery; while they were much slower than adults, they did show the linear rate of rotation reported by Shepard and Metzler. However, Dean and Harvey (1979), using a similar paradigm with

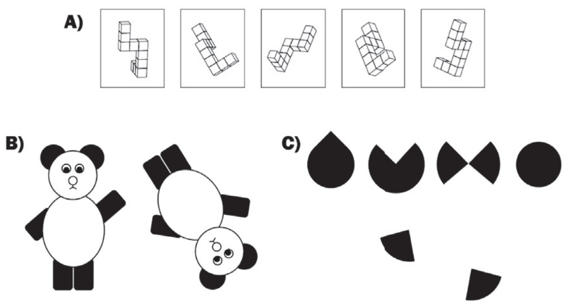

Fig. 1 Author renditions of examples of mental rotation tests that use different questions, dimensionality, and stimuli. Note. *A) Example of an adult mental rotation test based on Vandenburg and Kuse, 1978 (as described in Wilbur et al., 2024). B) Marmor (1975, 1977) bears mental rotation. C) Levine et al. (1999) Children's Mental Transformation Test.*

different stimuli, failed to replicate these findings, ultimately suggesting that children could not successfully perform mental rotation until age 6. Since then, numerous studies on children's mental rotation have often yielded conflicting results, particularly regarding the age at which mental rotation skills develop. For example, some studies suggest that Marmor's original findings were accurate and that children develop mental rotation skills around 4.5 years old (i.e., Frick et al., 2013), while more recent studies indicate that even younger children exhibit mental rotation skills (i.e., Krüger, 2018). These findings are not only contradictory within this age range, but are also contradictory with infant studies of mental rotation that show infants as young as three-months-old show precursors of mental rotation, such as detecting mirror reflections at various orientations (Enge et al., 2023; Moore & Johnson, 2008; Quinn & Liben, 2008). When considering the entire literature onmental rotation from birth to early childhood, there are unanswered questions about the developmental trajectory of mental rotation (Frick et al., 2014a). Further, the apparent dip in mental rotation ability of pre-school aged children may further suggest potential issues with the mental rotation tests used with this age range; these tests may be too confusing or have too high of cognitive demands.

2.2 Perspective taking

Perspective taking is another common category of spatial thinking tests; however, there is a significant gap between the use of mental rotation and

Table 2 Examples of tests used to assess children's perspective taking.

Test name	Source	Original age range (years old)
Short Object Perspective-Taking Task (sOPT)	Kozhevnikov and Hegarty, 2001	Undergraduates (also used with children 8- to 16-years-old, e.g. (Orefice et al., 2024))
Perspective-Taking Test for Children (PTT-C)	Frick et al., 2014b	4–8
Photographic Perspective Taking Test	Liben and Downs, 1993	5.5–11.5

perspective taking (Uttal et al., 2024). Spatial perspective taking is generally defined as mentally adopting a viewpoint different from one's own. The literature on spatial perspective taking is extensive, as "perspective taking" can refer to various abilities, not all of which are spatial. For example, perspective taking can involve understanding another person's beliefs or feelings, as well as viewing an environment or object from a different angle (Healey & Grossman, 2018; Michelon & Zacks, 2006; Tarampi et al., 2016). However, only the latter interpretation of perspective taking is typically considered a spatial skill (Brucato, 2022). Although some hypotheses suggest that all types of perspective taking share common cognitive processes (Parkinson & Wheatley, 2013), recent work has supported the claim that spatial perspective taking is a distinct process (Brucato, 2022).

Unlike mental rotation, assessments of spatial perspective taking have historically focused more on young children. One of the first spatial perspective taking assessments is the Three Mountains Test developed by Piaget and Inhelder (1956). In this test, a child is seated next to a model of three distinct mountains, while a doll is positioned differently around the Table. The child is then asked to identify the view of the mountains that the doll would see. The cognitive process involved in perspective taking requires imagining a different frame of reference. Research has shown that perspective taking is distinct from mental rotation in both adults and children (Frick & Pichelmann, 2023; Hegarty & Waller, 2004). Furthermore, like mental rotation tests, there are various versions of perspective taking tests for children, though there are fewer of them and the differences between these tests are not as pronounced.

2.3 Block design test

While mental rotation and perspective-taking tests are designed to target a specific type of spatial thinking skills, some tests are used as more general spatial thinking assessments. One such general spatial thinking test is the Block Design Test (Groth-Marnat & Teal, 2000; Jirout & Newcombe, 2015). In this test, participants must recreate a design using colored blocks within a certain time limit. The Block Design Test was not originally intended to be a spatial thinking test but rather a general intelligence test to be used primarily for clinical purposes (Hutt, 1932; Kohs, 1920). Since it was first developed, researchers have used the block design test in other settings, particularly as a spatial assessment in research and educational settings (McKee et al., 2024). Despite not being originally designed as a spatial thinking assessment, research indicates that performance on the Block Design Test is correlated with other spatial assessments and experiences (Caldera et al., 1999; Casey et al., 2008). Because the Block Design Test was not designed to target a specific spatial skill, its use in spatial thinking research is varied; researchers in different contexts have referred to the Block Design Test as an assessment of spatial visualization, visual processing, spatial relations, and general visuo-spatial ability (Casey et al., 2008; Flanagan & Kaufman, 2004; Jirout & Newcombe, 2015). Further, the Block Design Test also exemplifies spatial construction assessments, a type of spatial assessment primarily used with young children (Verdine et al., 2014). These spatial construction or block construction assessments typically aim to mimic naturalistic block-building play.

2.4 Other small-scale spatial tests

Outside of mental rotation and perspective taking, there are very few categories of spatial thinking with multiple assessments available for young children. For instance, several spatial skills have multiple assessments for adults, such as cross-sectioning (Cohen & Hegarty, 2007; Ormand et al., 2014, 2017; Titus & Horsman, 2009), but there are far fewer cross-sectioning assessments specifically designed for children (Ratliff et al., 2010). This lack of established assessments does not indicate a complete absence of research on these spatial skills; rather, these skills have often been measured in children using one-off, ad hoc tests. Additionally, some assessments originally developed for adults have been adapted for use with children (Aguilar Ramirez et al., 2021; Gori et al., 2024; Hoyek et al., 2012; Morawietz et al., 2024).

Table 3 Examples of other small-scale spatial assessments for young children.

Test name	Source	Original age range (years old)
Crossing Sectioning for Children	Ratliff et al., 2010	5–9
Spatial Scaling Task	Frick and Newcombe 2012	3–6
Diagrammatic Representations Task	Frick and Newcombe, 2015	4–8
Proportional Reasoning	Möhring et al., 2015	4–5
Mental Folding Task for Children	Harris et al., 2013	4–7
Test of Spatial Ability (TOSA) 2-D & 3-D	Verdine, Golinkoff et al., 2014, 2014	3–5
Block Design Test	WISC Version: Weschler, 2014	6–16

Furthermore, the scarcity of tests for certain spatial skills can be beneficial in research if it leads to multiple studies using the same test. This reduction in variation can enhance the comparability of results across different studies, as performance in various samples on the same assessment can be directly compared. For example, the Children's Embedded Figures Task (Witkin et al., 1971, 1971) has been utilized by various researchers in studies of children's spatial thinking over many decades, including those that also examined children's STEM abilities (Hodgkiss et al., 2018; Tracy, 1987; Quaiser-Pohl et al., 2004). However, many of the spatial assessments currently in use were developed only in the last decade.

2.5 Large scale spatial assessments

There are more spatial assessments for small-scale than for large-scale spatial thinking, both for adults and children (Uttal et al., 2024). The relative lack of large-scale assessments is at least partly due to the challenges of testing large-scale spatial skills. Navigation is often assessed using virtual environments, sometimes presented through virtual reality (Nguyen et al., 2023). To run these simulations effectively, researchers need suitable computers, screens, and headsets (if using virtual reality). Furthermore, creating and sharing virtual environments is more challenging than sharing a paper-and-pencil assessment.

Table 4 Examples of test used to assess children's navigation.

Test name	Source	Original age range (years old)
Silcton	Schinazi et al., 2013	Various; 8–16 (Nazareth et al., 2018)
Virtual Path Mazes	Jansen-Osmann and Wiedenbauer, 2004	7–8

3. Challenges of children's spatial assessments

With the increased interest in spatial thinking, many researchers in education and related fields have begun using spatial assessments in their research. However, it can be challenging for those with limited knowledge of spatial cognition to find, select, and administer these assessments (Schneck & Nathan, 2024). Additionally, there are broader issues related to the absence of a comprehensive framework for spatial thinking as a whole (see Uttal et al., 2024 for a review). These issues also apply to assessments for children. However, creating assessments for children, especially young ones, presents unique challenges that can compound the general difficulties associated with spatial assessments, resulting in children's spatial assessments being significantly less effective than those for adults. In this section, we review these unique challenges.

3.1 Challenges of designing assessments understandable for children

Perhaps the main challenge in creating assessments for children is balancing the need to capture the construct of interest while keeping the design simple enough for children to understand the task. Despite numerous cognitive and attentional differences between adults and children, many assessments for children have been developed by modifying adult tests, changing only a few features, such as the stimuli or total number of items. For example, in one of the first mental rotation tests for children, the Shepard & Metzler (1971) mental rotation paradigm was adapted, replacing the three-dimensional cube figures. with child-friendly drawings of bears and ice cream cones (Marmor, 1975, 1977). Similarly, the Animal-MRT maintained the general structure of the Vandenburg & Kuse (1978) test but substituted the cube figures. with pictures of animals. Although these changes make tests more child-friendly compared to the adult tests, researchers have more recently rethought how mental rotation questions can be asked in more child-friendly ways in new test paradigms (i.e. Beckner et al., 2023; Frick et al., 2013; Pedrett et al., 2023).

Effects of Training and Instruction. Researchers have sought to ensure that children understand assessments by including more practice and training questions. However, it is widely accepted that children's spatial skills are malleable (Yang et al., 2020). A meta-analysis of spatial training found that improvement can occur within a single session (Uttal et al., 2013). Thus, the question arises at what point additional practice and training questions shift from demonstrating what children should do on a task to training the targeted skill. Platt and Cohen (1981) demonstrated significant performance differences between children who received training and those who did not on a mental rotation test. Moreover, the specific kind of instruction and practice children receive can also affect their performance. Studies have shown that physical experience with manipulatable stimuli can enhance children's spatial thinking and can be used as instructional material for assessments (Byrne et al., 2023; Frick et al., 2013a; Frick et al., 2013b; Ha & Fang, 2018). Further, the language used in the instructions of a task may influence how children approach or understand the task. For example, in a study done with adults, the language used in a navigation task affected how participants completed the task (Boone, Maghen, & Hegarty, 2019). Thus, if the instructions of a task give explicit instruction related to how participants should solve a task, the participants might use strategies they otherwise would not. Although these tools can be used to explain a task to children, they might also give children too much information about how to solve a particular task, leading to artificially improved performance.

Attentional differences between children and adults. Another challenge in designing tests for children, compared to adults, is their lower attentional capacities (Betts et al., 2006). Tests for children are typically shorter than those for adults, and those for younger children are generally shorter than those for older children. Differences in the number of items in a test can affect its reliability, a point discussed later. Additionally, tests for children often use engaging designs to maintain their interest throughout the assessment. However, design choices intended to enhance engagement may interfere with or distract children from completing the task. An example of tests with conflicting designs is the Picture Rotation Test and the Ghost Test (Frick et al., 2013; Quaiser-Pohl, 2003). In the Picture Rotation Test, the stimuli consist of colorful pictures of animals, selected after pilot testing indicated that children were more motivated by colored images than by black-and-white stimuli.

In contrast, the ghost stimuli in the Ghost Test are grayscale and lack many details, which reduces potential distractions for children.

Individual differences among children. Like any population, children exhibit individual variation. Differences among children can lead to some successfully completing an assessment while others of the same age perform at chance level (i.e., Frick et al., 2013). A more important variation to consider is the differences in children's approaches to assessments. For example, consider research on adults' varying strategy use on mental rotation tests. Studies showing that adults employ non-rotational strategies on these tests have existed for decades (Boone & Hegarty, 2017; Hegarty, 2018; Just & Carpenter, 1985). Although research on adult strategy use in these tests is extensive, relatively little has focused on how children solve mental rotation problems, even though children have demonstrated the use of strategies other than rotation (Estes, 1998). Furthermore, researchers have suggested that children's inability to complete mental rotation tests successfully at young ages can stem from using inefficient strategies. For instance, Quaiser-Pohl et al. (2010) found that young children employed various strategies on the Picture Rotation Test, some more appropriate than others. In this test, children view a target picture of an animal or human and are asked to choose which of three answer choices matches the target. The two incorrect choices are mirror-reflected versions of the target image, and all three choices are rotated by varying amounts. One semi-appropriate strategy children used was selecting the answer choice with the smallest angular disparity. In other words, children who employed this strategy would choose the answer with the picture closest to upright, regardless of whether it matched the target image.

This difference in strategy use among children also indicates that children may perform differently on various tests. The least angular disparity strategy is not applicable to all types of mental rotation tests, as not all tests provide answer choices that vary in degrees of rotation. Additionally, without understanding how a child solved a particular assessment, it is challenging to determine whether poor performance was due to an inability to demonstrate the targeted skill or a lack of understanding of the task. Young children may not grasp what is being asked of them on a given assessment, which could contribute to their failure on mental transformation tests, despite infants showing precursors to mental transformation (Frick et al., 2014a).

Many children around the ages of 3–5 likely have not encountered judgment-based multiple-choice tests before. In educational settings, policy recommendations suggest that more formal test scores should be accompanied by informal observations due to children's unpredictability on tests (Epstein et al., 2004). Furthermore, there is even less research on children's strategy use or response patterns on non-mental rotation spatial assessments.

3.2 Test availability

One major limitation in children's spatial assessment is the availability of tests. Although this issue is also a concern for adult spatial tests (Uttal et al., 2024), children's assessments face a unique challenge: there are both too many and too few tests. Many tests have been created, but only afew are widely accessible to other researchers.

Overabundance of tests. One characteristic of spatial assessment is that tests are often developed for use in one or two studies due to reasons such as lack of access to or funding for existing tests (Uttal et al., 2024). This results not only in an overabundance of tests but also in less comparable results across studies. Furthermore, when spatial tests are developed for a study, they are typically mental rotation tests. A notable feature of the literature on children's mental rotation is the existence of numerous different mental rotation tests. These tests can vary in several ways, including their stimuli, the number of practice questions, and the number of answer choices. Additional differences among these tests include the type of questions, which means many children's mental rotation tests do not resemble the original mental rotation tests (Frick & Pichelmann, 2023). The presence and nature of alternative strategies used by adults have been shown to differ depending on specific task characteristics of the mental rotation test (Boone & Hegarty, 2017; Jost & Jansen, 2024). Thus, specific test characteristics that vary across tests, such as stimuli or question type, may influence how individuals approach the problem. Different mental rotation tests for children may measure different constructs due to these design differences, leading individuals to solve the tests in various ways, possibly using non-spatial strategies. Like the potential use of alternative strategies by children, the effects of different test characteristics on children's performance have not been systematically evaluated.

Lack of availability of children's tests. While many tests have been developed to measure children's spatial thinking, most are not widely accessible. Currently, there are no online resources that specifically provide free spatial assessments for children or information on where to find various tests. The Spatial Intelligence Learning Center (SILC; spatiallearning.org) maintains a repository of spatial assessments that includes some for children. However, out of the 41 resources listed, only seven assessments are explicitly labeled as children's assessments in their name or description (excluding surveys and coding manuals). Of these seven, SILC has complete test materials for only two: the Children's Mental Transformation Test and the Mental Folding Test for Children. Consequently, the Children's Mental Transformation Test is one of the most used assessments for children.

Furthermore, not all children's tests can be easily shared through online resources because they require physical manipulatives, either in the instructions or as part of the test itself (i.e., Casey et al., 2008; Frick et al., 2013). One example of a physical assessment is the Test of Spatial Assembly (TOSA), which asks children to recreate a target design using either flat geometric pieces (2D) or interlocking blocks (3D) (Verdine et al., 2014). To administer this test, a researcher must obtain the target stimuli designs, scoring manual, and purchase the physical manipulatives needed for the test. These physical assessments create additional barriers to administration, and researchers cannot use them for online studies, which have become increasingly common post-COVID-19 (Lapidow et al., 2021). The lack of widely available tests fuels ad-hoc test creation, leading to a cycle where the unavailability of existing tests results in the over-creation of new ones.

Modifications of tests. Sometimes researchers do not create entirely new tests but modify existing ones. These changes can range from seemingly minor adjustments, such as reducing the total number of items or digitizing a previous paper-and-pencil test, to more significant alterations, like developing a new set of stimuli based on an existing test. Similar to our uncertainty about how many tests of the same spatial skill relate to each other, we do not know how these modifications influence children's understanding, performance, and strategy use on different tests. Furthermore, altering one test characteristic may affect other characteristics. For example, Bambha and Casasola (2021) report on digitizing the Picture Rotation Test, a mental rotation test traditionally administered in person, for use in an online study. This change not only alters the modality of the test but also requires updating the instructions to reflect that the experimenter is not in the same room as the child.

Missing Spatial Assessments. The types of spatial thinking assessments used with children tend to mirror those used with adults. In other words, the spatial constructs measured in adults are also assessed in children. While this overlap can be useful for researching the development of specific spatial skills, the focus on a limited set of abilities often leads to the oversight or complete neglect of important spatial skills (Uttal et al., 2024). For example, non-rigid spatial skills, such as those used when mentally bending or breaking an object, are considered particularly important for STEM capability, yet they are rarely, if ever, measured in young children (Atit et al., 2013, 2020; Resnick & Shipley, 2013). Additionally, there may be areas of spatial thinking that have not yet been identified in either adults or children.

3.3 Lack of psychometric rigor

Even in cases where established children's spatial assessments exist, these tests can lack adequate standardization and psychometric evaluation. Many researchers do not report the psychometric properties of their tests. For example, when Levine et al. (1999) introduced the Children's Mental Transformation Task, they did not provide reliability measures or discuss validation processes for the test. Furthermore, the reported reliability of some children's tests has varied substantially across studies; a test may demonstrate acceptable reliability in one study but low reliability in another. For example, the reported Cronbach's alpha for different forms of the Children's Mental Transformation Test has ranged from .24 to .82 (Casasola et al., 2020; Ramírez, 2018). Some of these problems may arise in part because children's spatial assessments may be inherently more challenging to evaluate than adult tests.

Reliability. The reliability of an assessment is a fundamental aspect of determining its quality (Schmidt & Hunter, 1996). The most used measure of test reliability among psychologists is Cronbach's α, although some psychometricians have pointed out its limitations (Revelle & Condon, 2019). However, calculating traditional reliability measures, such as α or Guttman's λ_2, which rely on inter-item correlations, can be misleading or challenging to compute for some children's tests. For example, tests with items that vary in difficulty can result in decreased reliability under classical test theory (Gulliksen, 1945; Hambleton & Jones, 1993). Many spatial tests exhibit variation in item difficulty, such as differing angular disparities in mental rotation tests or varying amounts of folds in mental folding tests (Frick, 2019; Harris et al., 2013). Additionally, some children's tests are not paper-and-pencil based but instead require a child to perform a physical task (e.g., Casey et al., 2008; Verdine, Golinkoff, et al., 2014). These tests can resonate better with children because they mimic natural activities, like block-building, rather than relying on multiple-choice questions. Although the reliability of these tests can be calculated by simplifying a child's response to correct or incorrect, this scoring method overlooks the richness that can be captured through more qualitative approaches. Moreover, even with quantitative scoring systems, these physical tests have shown low reliability in some studies (Dearing et al., 2012). Thus, a significant challenge in designing children's spatial assessments lies in balancing the creation of open-ended, qualitative assessments that children may understand better with the need for assessments rooted in traditional psychometric rigor.

Another factor that can influence test reliability calculations is sample size (Charter, 1999). Studies involving children tend to have fewer participants than those involving adults for various reasons, such as the difficulty of recruitment (Hurwitz et al., 2017). Consequently, if measures like Cronbach's α or Guttman's λ_2 are calculated for a children's assessment using a small sample size, the resulting reliability may not be accurate. Collecting large samples of children can also be quite expensive, as tests that do collect a large sample to create testing norms, such as the Block Design Test, tend to be costly to acquire. Furthermore, more complex psychometric models, such as Item Response Theory (IRT), also require large sample sizes, typically in the hundreds, to be effective (Edelen & Reeve, 2007). For example, Munns et al. (2023) recently conducted IRT analysis on four different adult mental rotation tests. For this analysis, they obtained a large, representative sample of 500 adults recruited online through Qualtrics. Collecting a similarly sized, nationally representative sample of children would be much more challenging, as most widely used online data collection tools are primarily suited for gathering data from adult participants. Despite recent pushes for more rigorous psychometric evaluations of spatial assessments, children's assessments have so far not been included in this work (Brucato et al., 2023; Munns et al., 2023; Uttal et al., 2024).

Validity. In addition to reliability, proper validation of tests is a core aspect of psychometric work. As mentioned earlier in the discussion of challenges in designing spatial assessments for children, creating a test that is both age-appropriate and accurately measures the target construct is challenging. Therefore, validating children's assessments is crucial to ensure the target construct is being assessed. Many children's assessments are either not validated, or the validation procedures are not reported. Furthermore, validating assessments for children is challenging since one of the most common measures researchers use to validate assessments is convergent validity. Convergent validity suggests that an assessment is valid if it relates to an existing assessment of the same or similar ability (DeVon et al., 2007). For example, if a researcher creates a new assessment for children on mental folding, they might administer both the new test and a mental rotation test. If the children's performance on the two spatial tests is correlated, this suggests the new test is measuring a facet of spatial thinking. However, conducting this kind of validation depends on the validity of the comparison test; if there are no gold-standard assessments to compare new tests to, validating new tests can be difficult. Thus, the lack of psychometrically validated spatial assessments for children can result in more assessments not being properly validated.

4. Interim summary

Through the previous sections, we have established the importance of researching the development of spatial thinking. However, despite the attention that spatial thinking has received in the last few decades as a potential tool to improve students' STEM learning, the current state of spatial assessments for young children has room for improvement. Compared to tests for adults, several factors make it more challenging to create, share, and validate assessments for children. Although these challenges present hurdles to improving children's spatial assessments, they are not insurmountable. Additionally, researchers have been working on addressing these challenges and are making progress towards valid, age-appropriate spatial measures for children (i.e. Frick & Pichelmann, 2023). Nevertheless, the limitations of these assessments should be considered when evaluating research that has used them, especially in studies with mixed results or many unknowns.

5. Implications of children's spatial assessments on spatial-stem research

One area of research that relies on spatial assessments and has many open questions is the relationship between spatial thinking and STEM performance. The evidence linking spatial thinking to STEM outcomes is continuously increasing; however, there is still much unexplored potential in studying the *development* of this relationship (Newcombe & Frick, 2010).

Spatial thinking plays an important role in early STEM education. Evidence of its importance includes the integration of the skill set in elementary science and math education standards. For example, according to the Next Generation Science Standards, elementary school science education should emphasize skills such as interpreting and creating diagrams and developing physical models (NGSS, 2013). Similarly, Common Core math standards state that early math should include concepts like measurement, understanding a number line, and reasoning about shapes (NGACB & CCSSO, 2010). Although the specific topics of these standards differ across content areas (i.e., math versus science), the prevalence of spatial thinking underlines its substantive and foundational role in early STEM learning.

Furthermore, due to the importance of spatial thinking in early STEM mastery, spatial training interventions may be particularly effective for young children. In Uttal et al. (2013) meta-analysis of the malleability of spatial thinking, training with children under the age of 13 was shown to be effective compared to a control group, with a Hedge's $g = .61$. Years later, Yang et al., (2020) conducted a meta-analysis on spatial training specifically with children between the ages of 3 and 8, finding an overall Hedge's $g = .96$. While full interpretation of these results depends on understanding the heterogeneity within the included studies, their general positive findings suggest that spatial training with young children is effective. Experimental studies have also provided evidence that improvements in spatial ability in young children can transfer to STEM domains (Cheng & Mix, 2014; Mix et al., 2020); however, other studies have shown no transfer to STEM (Hawes et al., 2015).

Yet, as stated earlier, the results of spatial-STEM research with young children depend on the assessments that were used in the research. If an un-validated assessment was used, the results of the study should be interpreted with that in mind. Furthermore, assessments can play a larger role in research than merely a dependent variable. Thus, the state of spatial assessments can have consequences that extend beyond laboratory research. In this section, we discuss the potential implications of the current state of children's spatial assessment on both spatial-STEM research and educational contexts.

5.1 The role of assessment in spatial-stem research

While the quality of assessment is important in any research, it is particularly critical in spatial training and intervention studies. In most spatial training studies, one or two specific spatial skills are targeted, almost exclusively defined by spatial assessments. For example, multiple studies have examined the effects of mental rotation training on both spatial thinking and STEM skills in children (Cheng & Mix, 2014; Cheung et al., 2020; Fernández-Méndez et al., 2018; Hawes et al., 2015; Rodán et al., 2019). Although there is promising evidence that mental rotation-focused training can enhance mathematical abilities (Cheng & Mix, 2014), it is worth questioning whether the extensive focus on mental rotation stems from its true importance for STEM or from an overabundance and reliance on mental rotation assessments.

In a recent meta-analysis of the relationship between spatial skills and mathematics, of the 93 tests or test batteries used in the 45 included studies, approximately one-third were mental rotation tests or included mental rotation questions (Atit et al., 2022). No other spatial skill was as prominently represented as mental rotation among these tests. In a different meta-analysis specific to training spatial thinking in young children, mental rotation accounted for about 40 percent of the spatial outcome measures reported in the included studies (Yang et al., 2020). Thus, current theories regarding how spatial skills contribute to STEM abilities depend on a limited number of spatial thinking tests. Furthermore, without a wide range of spatial assessments for young children, researchers are unable to determine which spatial skills are related to specific STEM abilities.

One of the main motivators for researching spatial thinking and STEM is the potential to leverage the malleable nature of spatial thinking to improve students' spatial skills, with the hope that this improvement transfers to STEM ability. Spatial assessments also play an important role in spatial training studies. Not only do they serve as outcome measures in intervention studies, but they can also inform the training conducted during the study. For example, in Cheng & Mix (2014)'s spatial intervention study, the training children received was mental rotation training based on the Children's Mental Transformation Test. Bower et al., (2020)'s spatial training was based on the Test of Spatial Ability. This type of assessment-targeted training can lead to near transfer; children trained on a specific assessment can show improvement on that assessment. However, this kind of assessment-based training also limits the potential for transfer to various STEM domains; far transfer from spatial training to STEM would depend on the strength of the relation between the assessment informing the training and the specific STEM measure of interest.

5.2 Educational implications and outcomes

The state of spatial assessments for children can also impact education. Spatial thinking has historically been overlooked in traditional educational settings (Gagnier & Fisher, 2016; Mathewson, 1999). Unvalidated spatial assessments for children used in spatial-STEM research may obscure study results, making it difficult to present compelling arguments to education stakeholders about the educational importance of spatial thinking. Mixed research results also impede progress in evaluating best practices for leveraging spatial thinking in STEM education. Furthermore, the lack of widely available and validated spatial thinking assessments for children affects the diversity of students pursuing STEM in the future. The current limitations associated with spatial assessments make it challenging to identify students who could benefit from spatial training to enhance their STEM performance. Additionally, it restricts the identification of students who excel in spatial thinking and may be more likely to succeed in STEM (Lakin & Wai, 2020; Veurink & Sorby, 2019; Wai & Kell, 2017; Wai & Lakin, 2020). Identifying students from disadvantaged backgrounds who excel in spatial thinking can foster their potential and increase representation in STEM from historically underrepresented groups (Wai & Lakin, 2020).

6. Recommendations for future research

In this chapter, we reviewed the current state of children's spatial assessments, the challenges in designing these assessments, and their implications for broader research. While recognizing the existing problems with children's spatial assessments is the first step toward improvement, more dedicated efforts are necessary to achieve significant enhancements. This work is also essential for advancing research in spatial thinking, including the reevaluation of conceptual frameworks of spatial thinking (Uttal et al., 2024). Therefore, in this section, we offer recommendations to improve the state of children's spatial assessments.

6.1 Research focused on children's tests

One of the primary recommendations for improving children's assessments is to conduct research aimed at understanding these assessments and their properties. While there is significant momentum in evaluating adult spatial assessments through item response theory-based studies (Brucato et al., 2023; Munns et al., 2023), we recommend that research on children's assessments should begin by understanding existing assessments before attempting to refine them.

Systematically review children's assessments. To understand these assessments, researchers must know what assessments exist and have been used with young children. This work could help clarify how to locate different spatial assessments and lead to understanding differences in how children perform on various tests, particularly of the same construct. For example, Nguyen et al. (2023) recently reviewed the use of virtual environments to assess the development of navigation in children and adolescents. Their review, while not a full systematic review, compares the paradigms of different types of virtual environment studies, which is useful for understanding which aspects of navigation are and are not captured by these assessments. Furthermore, these kinds of reviews also highlight age ranges that are lacking assessments for certain types of spatial skills.

Effects of test design. Researchers should examine how various test characteristics affect children's performance and strategy use. By comparing different tests and versions of the same test, researchers can determine the significance of specific test characteristics. For instance, studies have explored the effects of modality differences on children's performance between paper-and-pencil and computerized versions of the same test (Bambha & Casasola, 2021; Frick & Pichelmann, 2023), as well as between paper-and-pencil tests and physically manipulable versions of the same test (Frick et al., 2013). Additionally, research has investigated the impact of different types of stimuli, such as variations between animals and cubes (Neuburger et al., 2011). However, many test characteristics remain unstudied, such as the spatial arrangement of questions and answer choices. Moreover, most of the research on test characteristics conducted thus far has focused on mental rotation tests.

Children's strategy use. To determine if a particular test is measuring the desired construct, it is important to examine how children approach the test. For example, understanding children's erroneous strategy use (or in other words, children failing to solve a task as intended) can provide insight into the test's validity and indicate whether changes are necessary to limit unwanted strategies. Additionally, analyzing children's strategy use over time can shed light on the development of spatial thinking skills (Quaiser-Pohl et al., 2010).

6.2 Make assessments accessible

To limit the number of tests created ad-hoc or modified for individual studies, reliable, validated assessments should be easily accessible to researchers. This accessibility includes not only making assessments available but also ensuring their formats of assessments are user-friendly by researchers.

Synthesize test resources and information. In addition to systematically reviewing existing assessments, efforts should be made to create a synthesized collection of test information. This collection should detail what each test measures, how to obtain it, how to administer it, its psychometric properties, and the appropriate age ranges for its use.

Adapt assessments for online studies. Many researchers rely on online studies to recruit participants and conduct research with children. Online child testing platforms, such as LookIt, have facilitated this research possible for both synchronous and asynchronous studies (Scott & Schulz, 2017). Although promising research indicates that digitized versions of assessments perform similarly to in-person assessments, more standardization is needed for online spatial assessments (Bambha & Casasola, 2021). These platforms enable researchers to engage families they might otherwise be unable to reach in a lab setting and provide a shared database of families among researchers.

6.3 Diversify which spatial skills are assessed

A significant portion of research on spatial thinking is focused on mental rotation. This emphasis is prevalent in both spatial cognition research and in the operationalization of spatial thinking within various STEM domains. While some studies suggest that mental rotation assessments are uniquely related to certain STEM fields (Bates et al., 2021), the focus on mental rotation has been cyclical; promising results from these assessments lead to increased interest, generating more results related to mental rotation. Importantly, this focus does not imply that other spatial skills lack promising results. Children's perspective-taking and spatial assembly skills have both been shown to be malleable and related to specific areas of mathematics and science (Bower et al., 2020; Bower & Liben, 2021; Gilligan-Lee et al., 2023; Plummer et al., 2016; Verdine, Irwin, et al., 2014). However, the full potential of non-mental-rotation spatial skills has yet to be explored in children. Future research should diversify the spatial assessments used to inform interventions in spatial-STEM studies. Additionally, incorporating multiple spatial assessments within a study examining spatial and STEM abilities can help researchers understand whether the relationship between these skills is more general or specific. For example, what spatial skills, besides mental rotation, are related to specific mathematical concepts (e.g., number line comprehension)? To answer these questions, researchers must have access to valid spatial thinking assessments across a range of spatial skills.

6.4 Incorporate existing research from other cognitive assessments

Spatial cognition is not the only field that has developed cognitive assessments for children. However, research across different fields tends to be siloed. Therefore, researchers aiming to improve or create spatial assessments for children should not only examine existing spatial assessments but also learn from successful assessments in other domains. Various assessments have been developed within psychology to evaluate young children's executive functioning, fluid reasoning, analogical thinking, and other cognitive abilities (Anderson, 2002; Shivaram et al., 2023; Weschler, 2003). Additionally, cognitive skills related to spatial thinking, such as patterning skills, have been effectively assessed in children (Rittle-Johnson et al., 2013). Researchers in these cognitive areas are also advocating for improved assessments for young children. For instance, a recent study explored the gamification of an executive function assessment for preschool-aged children (Eng et al., 2024). By evaluating which types of assessments have been shown to be valid and reliable in other cognitive domains, researchers can apply similar design practices to create and refine spatial assessments for children.

7. Conclusions

Spatial thinking research is an important and growing field, with increasing evidence suggesting that spatial thinking interventions can benefit students across various educational contexts (Gilligan-Lee et al., 2022; Newcombe, 2017). Although this research has been limited by unreliable, unvalidated, inaccessible, or nonexistent spatial assessments for children, these challenges can be addressed. Some researchers have begun focusing on validating children's spatial assessments and providing psychometric information (Frick & Pichelmann, 2023). Furthermore, this work parallels similar studies on adult spatial assessments (Brucato et al., 2023; Munns et al., 2023; Uttal et al., 2024). Given the crucial role of assessment in research, this assessment-focused work is foundational for improving theories and frameworks of spatial cognition, understanding the relationship between spatial thinking and STEM proficiency, and developing effective spatial interventions.

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